

$$Q1) \hat{y} = \text{Proj}_W y ; y = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, W = \text{span} \left\{ \underset{u_1}{\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}}, \underset{u_2}{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}} \right\}$$

Note u_1 and u_2 are orthogonal.

$$\therefore \hat{y} = \frac{u_1 \cdot y}{u_1 \cdot u_1} u_1 + \frac{u_2 \cdot y}{u_2 \cdot u_2} u_2$$

$$= \left(\frac{2}{2}\right) \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + \left(\frac{2}{1}\right) \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Ans: (d) $\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$

$$Q2) A = \begin{bmatrix} 1 & -2 \\ 0 & 0 \\ 2 & -4 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$\begin{matrix} u_1 & u_2 \end{matrix}$

Notice that $u_2 = -2u_1 \Rightarrow$ dependent col.

$\therefore$ only 1 col is needed.

$$\therefore \hat{y} = \left(\frac{u_1 \cdot b}{u_1 \cdot u_1}\right) u_1 = \left(\frac{7}{5}\right) \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 1.4 \\ 0 \\ 2.8 \end{bmatrix}$$

$$\therefore x_p = \begin{bmatrix} 7/5 \\ 0 \end{bmatrix} = \text{particular.}$$

$$x_{\text{hom}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \lambda ; \lambda \in \mathbb{R}.$$

$$\therefore x_{\text{general}} = \begin{bmatrix} 7/5 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Ans:

$$\hat{x} = \begin{bmatrix} 3.4 \\ 1 \end{bmatrix}$$

using $\lambda = 1$

Q3(a) $\hat{x}$ is not unique, when $\text{col}(A)$ are dependent only unique when $(A^T A)^{-1}$ exist

(False)

3(b) $\text{Proj}_W y$ is always unique. (True)

3(c) ~~True~~ : if $(A) \overset{\text{span}}{=} (B)$.

(C) then $\text{Proj}_A y = \text{Proj}_B y$.

But not enough information given in question.

4(a) Normal eqⁿ will always have a solⁿ.

(False)

4(b) (True) : P_1 generated from $QQ^T = \text{same.}$

$$P_2 = A(A^T A)^{-1} A^T$$

$\underbrace{\hspace{10em}}_{\hat{x}}$
P.

Q5) $A = \begin{bmatrix} 1 & 0 \\ 1 & 2 \\ 0 & 2 \end{bmatrix}$; $P = A(A^T A)^{-1} A^T$

$$= \frac{1}{3} \begin{bmatrix} 2 & 1 & -1 \\ 1 & 2 & 1 \\ -1 & 1 & 2 \end{bmatrix} *$$

Ans = (A)

Q6) $z = \left(\frac{2e^{j\pi/4}}{1-j} \right)^{1/2}$

$$= \left(\frac{2 \angle \pi/4}{\sqrt{2} \angle -\pi/4} \right)^{1/2} = \left(\sqrt{2} \angle \frac{\pi}{2} \right)^{1/2}$$

$$= \left(\sqrt{2} \right) \angle \frac{\frac{\pi}{2} + k(2\pi)}{2}$$

$$= 1.189 \angle \frac{\pi}{4}, \frac{5\pi}{4}$$

Ans

(B) $1.19 e^{j\pi/4}$

Q7) row 2 of $W = \left[\leftarrow e^{-j\frac{2\pi}{8}(2)(n)} \rightarrow \right]$ 1 2

$n = 0 \dots 7$

$$= \left[\leftarrow e^{-j\frac{\pi}{2}n} \rightarrow \right]$$

$\therefore$ rotates clockwise at $\frac{\pi}{2}$ rad/sample

Q8) $X(k) = 4 \begin{bmatrix} 1 \\ 0 \\ 3+j \\ 0 \end{bmatrix}$

$$x(n) = \left(\frac{1}{N} W^{-1} \right) X(k)$$

$$= \begin{bmatrix} 4+j \\ -2-j \\ 4+j \\ -2-j \end{bmatrix}$$

Ans
(A)

Q9) Given $A = \begin{bmatrix} 2 & 2 \\ 2 & -1 \end{bmatrix}$.

$(E, D) = \text{eig}(A)$

$E = \begin{bmatrix} 1 & -2 \\ -2 & -1 \end{bmatrix}, D = \begin{bmatrix} -2 & 0 \\ 0 & 3 \end{bmatrix}$.

$\uparrow \quad \quad \uparrow$
 $v_1 \quad v_2$

$\lambda_1 = -2, \lambda_2 = 3$

Ans: **False**: The eigen values are repeated
 (C)

Q10) $A = \begin{bmatrix} 4.2 & -3.8 & -2.2 \\ 0 & -3.9 & -3.6 \\ 0 & 5.4 & 5.1 \end{bmatrix}$

To solve find $\lambda_1, \lambda_2, \lambda_3$ for v_1, v_2, v_3

$= 0.5, -0.3, 1.5$

$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix}$

$\therefore x = 1v_1 + 2v_2 + \frac{1}{2}v_3$

$\therefore \lambda_3, v_3$ dominates.

$\therefore A^k (1v_1 + 2v_2 + \frac{1}{2}v_3)$

$= \underbrace{A^k v_1}_{\rightarrow 0} + 2A^k v_2 + \frac{1}{2}A^k v_3$

$\rightarrow 0 + \frac{1}{2} \lambda^k \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix}$

$= + \frac{1}{2} \left(\frac{3}{2}\right)^{k \rightarrow \infty} \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix}$

$= \begin{bmatrix} -\infty \\ +\infty \\ -\infty \end{bmatrix} \# \text{ Ans} = \text{(C)}$